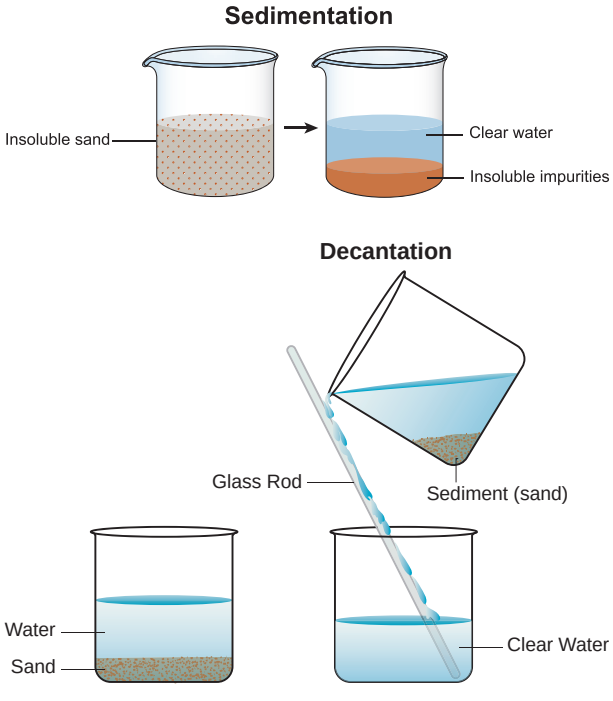
Sedimentation

Rice and pulses are often mixed with very fine straw, husk or dust particles which have to be removed before cooking. Are you familiar with the way this is done at home? To remove these particles rice or pulses are washed in water. The lighter impurities float while heavier rice grains sink to the bottom. This is called **sedimentation**. The water with the impurities is carefully poured down leaving clean rice at the bottom. This is called **decantation.**

Separating mud from muddy water

Muddy water is a mixture of very fine particles of soil in water. What will happen if muddy water is left undisturbed for some time? Mud being heavy will settle down at the bottom of the beaker and will form the sediment. Water forms the top layer and is called the supernatant liquid.



The settling down of heavier components of a mixture when allowed to remain undisturbed for some time is called sedimentation.

Decantation

This process is done after sedimentation. The supernatant liquid is slowly poured out from the container without disturbing the sediment. The part that settles down the bottom of the liquid is called sediment. The water that is obtained after decantation is called the decantate. The process of separating liquid above the sediment is called decantation.

But even after decantation the water is not completely free from fine soil particles. How can we remove this? We can do this by filtration. Do you think a strainer or a cloth can filter theses very fine particles? Do it by yourself and find out.

Filtration

We use filter papers to remove the finer impurities. A filter paper has very fine pores much smaller than soil particles. Let us see how to use the filter paper.

Take a piece of filter paper. Fold it to make a cone (see figure). Slowly pour the muddy water over the filter paper. On filtration clear water (filtrate) flows down the funnel and mud settles as residue on the filter paper. The method of separating insoluble component (sand, mud etc.) from a mixture using a filter paper is called filtration. The liquid which passes through the filter and comes down is called **filtrate** and the insoluble component left behind on the filter paper is called **residue**.